

Steven Wu (Cheng-Ju Wu)

cw3834@columbia.edu | (669) 288-4917 | [linkedin.com/in/steven-cj-wu/](https://www.linkedin.com/in/steven-cj-wu/) | Santa Clara, CA 95050

Education

Santa Clara University (SCU) | Santa Clara, California

September 2022 – June 2026

Bachelor of Science in Mechanical Engineering

- **Relevant Courses:** Composite Materials, Fluid Mechanics, Thermodynamics, Finite Element Theory and Applications, Engineering and Design with Applied Machine Learning, Computational Mechanics, Continuum Mechanics

Skills and Core Competencies

- Languages: Mandarin (Native), English (Fluent), German (Basic)
- Technical: SOLIDWORKS, Creo, MATLAB, 3D Printing, Laser Cutting, Pytorch, Python, C++, GD&T

Experiences

Undergraduate Research Assistant (Kuehler Undergraduate Research Grant Recipient)

June 2025 – Jan 2026

Santa Clara University GIFTED Lab | Santa Clara, California

- Developed a Physics-Informed Neural Network (PINN) framework for topology optimization, integrating manufacturing constraints for 3D printing to enhance design feasibility.
- Implemented a multi-network architecture coupling a density prediction network with a displacement field solver, improving compliance accuracy and convergence stability compared to standard SIMP-based methods
- Utilized PyTorch for neural architecture design and automatic differentiation; visualized material density evolution using Matplotlib and NumPy.

R&D Mechanical Engineering Intern

July 2024 – September 2024

SINBON Electronics Co., Ltd. | New Taipei City, Taiwan

- Designed and refined injection-molded plastic and sheet-metal components in Creo for OEM electronics projects, balancing manufacturability, structural durability, and cost.
- Created bills of materials and design documentation to support component sourcing, assembly, and production.
- Collaborated with electrical engineers and production personnel to resolve mechanical-integration and manufacturability issues.

Extracurriculars

SCU Bronco Air DBVF (Design-Build-Vertical-Flight)

June 2025 – June 2026

Structure Team

- Work with six-member team in the design and fabrication of an autonomous hexacopter for the VFS Design-Build-Vertical-Flight Competition.
- Design airframe components and motor arms in SOLIDWORKS and coordinate prototype manufacturing and assembly.
- Reduced structural weight by 47.4% through FEA-driven topology optimization while maintaining a factor of safety above 2.0 and maximum deflection below 1 mm.
- Integrate autonomous-flight hardware using a Raspberry Pi and CubePilot Cube Orange+.

SCU Robotics System Lab

January 2024 – June 2024

Vertical Profiler Team member

- Inspected vertical profilers following research deployments, diagnosed damaged components, and recreated replacement parts in SOLIDWORKS for 3D printing.
- Designed modular mounts for action cameras and sensors, supporting field reconfiguration and instrumentation of the profiler platforms.

SCU Bronco Racing (Formula Society of Automotive Engineers)

September 2022 – June 2024

Chassis Sub-Team Member

- Utilized SOLIDWORKS to design the side impact structure of the car, ensuring the driver's safety.
- Performed finite element analysis to assess the cockpit for structural integrity and weight reduction.